

EGLANTINE FONROSE

5th-year Audiovisual and Multimedia Engineering Student – INSA Hauts-de-France
Email: eglantine.fonrose@gmail.com | Phone: +33 7 82 06 81 57 | GitHub:github.com/eglantinefonrose

Seeking an internship of at least 5 months starting in February 2027

Professional Experience

INA (French National Audiovisual Institute) – Internship September 2025 – January 2026

- **Accessibility optimization:** Benchmarking of AI transcription models, comparative evaluation of integrating **Whisper** (transcription AI) into **FFmpeg** for real-time subtitle generation, with analysis of **performance/quality trade-offs** and recommendation of the optimal configuration.
- **Multi-stream synchronization algorithm:** design of a method for **temporal overlap detection** between consecutive video/audio files, combining **structural image analysis (SSIM)** and **spectral audio analysis (MFCC)**.
- **Streaming protocol study:** benchmarking of **HLS players (HLS.js vs Shaka Player)** and proof of concept for **dynamic fMP4 segment generation**.

Personal project

Groovy Morning FM – RLAC January 2025 – September 2026

Creation of an **on-demand live radio platform**. Researching solutions to **segment radio streams in real time** using **AI**.

GitHub repository : github.com/eglantinefonrose/GroovyMorning_RLAC

Detailed documentation : eglantinefonrose.github.io/GroovyMorning_RLAC7.articles/fr/0.introduction-generale.html

Technical architecture

- **Segmenter Layer (Python – Flask):** segmentation of the radio stream into **segments** via **real-time transcription** (STT engine **Whisper/Kyutai**) and detection, using language models (**LLM**), of the **precise moment** a segment begins.
- **Backend Layer (Java – Jetty) :** execution engine driving **FFmpeg** to capture audio in one-second **micro-segments** and dynamically assemble them into custom **HLS streams**, **synchronized** with the segmentation layer.
- **Client Layer (Android – Kotlin/Compose):** control and playback **interface** allowing the user to schedule their listening grid and precisely navigate (**seeking**) within recorded segments, via **ExoPlayer**.

Models explored during training

- **Audio models:** Machine Learning (**Random Forest** on MFCC/RMS), **fine-tuning** (Wav2Vec2, AST, BEATS with soft voting and temporal smoothing), **hybrid approach** (jingle detection + segment detection).
- **Text models:** Machine Learning (**Random Forest** on TF-IDF), **Deep Learning** (fine-tuning CamemBERT, hybrid CamemBERT + Bi-LSTM + CRF), **LLM approach** (few-shot prompting via Mistral/Qwen or DeepSeek/Claude API).

eCommerce Tailored Clothes App Mars 2020 – Juin 2023

- SwiftUI, Firebase, GCS (<https://github.com/eglantinefonrose/fonrose-ecommerce-mobileapp-ios-v2>)

Education

INSA Hauts-de-France September 2024 – Present

Engineering degree in **Audiovisual** and **Multimedia**

Training in the field of **audiovisual engineering**, covering **signal processing** (Fourier transform, DCT), **compression** (JPEG, MPEG, H.264), **transmission** (information encoding, modulation), and **programming** (C/C++ programming, 3D computer graphics, multimedia networks)

INSA Hauts-de-France September 2022 – June 2024

Integrated Preparatory Program

Technical Skills

Programming languages (practical): Java, Swift, Python
Programming languages (academic): C++, SQL, C
Frontend: SwiftUI (practical); Angular 17 (via Gemini CLI coding agent)
Backend: Google Cloud Platform (Firebase, GCS), Jersey / Apache Tomcat, Flask, Quartz
IA – Transcription and audio: OpenAI Whisper, Kyutai
IA – Language models: DeepSeek API
IA – Machine Learning: BERT, Random Forest
IA – MLOps platform: Weights & Biases (WandB)
IDE: SQLite, XCode, IntelliJ IDEA (with debugger)
Audiovisual software: FFMPEG, Matlab
Build tools: Git, Gradle, UV, Python VirtualEnv
Audiovisual skills: Compression algorithms (JPEG, PNG, MPEG), audiovisual signal processing, network

Languages

French: Native language, **English:** B2